

Figure 21.7 Small-signal model used to calculate the RHP zero for amplifiers in Fig. 22.1a–d.

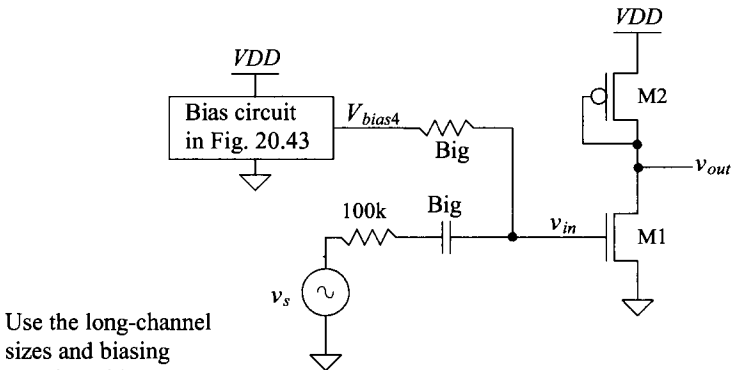
point is important. For the inverting amplifiers (which mean that at low frequencies their phase shift is 180°) in Fig. 21.1, for example, the input feeding directly to the output because of the RHP zero, without the inversion, can result in instability when the amplifier is used with feedback. The output can feed back and add to the input (positive feedback). The result is that the output of the amplifier will grow without limit (and, ultimately, the output of the amplifier will oscillate back and forth between some voltages). The magnitude response of the RHP zero is usually ignored because its value typically places it well beyond the pole frequency. However, in CMOS op-amp design, the phase shift of this RHP zero can, and will, result in stability and/or settling time issues.

Example 21.2

Determine the gain, magnitude, and phase shift of the amplifier shown in Fig. 21.8. Using an AC SPICE simulation, verify your hand calculations.

The bias circuit is used to bias the amplifier at the operating point seen in Table 9.1, that is, at a drain current of $20\ \mu\text{A}$. The big resistor and capacitor set the DC bias point but don't affect the AC operation of the circuit.

$$A_v = -\frac{\frac{1}{g_{m2}}}{\frac{1}{g_{m1}}} = -\frac{g_{m1}}{g_{m2}} = -\frac{150\ \mu\text{A}/V}{150\ \mu\text{A}/V} = -1$$



Use the long-channel sizes and biasing seen in Table 9.1.

Figure 21.8 Amplifier used in Ex. 21.2.

From Table 9.1, $C_{gs1} = 23.3 \text{ fF}$ and $C_{gd1} = 2 \text{ fF}$. The Miller input capacitance, because the gain of the amplifier is -1 , is then 4 fF . The input pole is located at

$$f_{in} = \frac{1}{2\pi \cdot R_s(C_{M1} + C_{gs1})} = \frac{1}{2\pi \cdot 100k \cdot (27.3 \text{ fF})} = 58 \text{ MHz} \quad (21.16)$$

From Table 9.1, $C_{sg2} = 70 \text{ fF}$. The Miller output capacitance is, again, 4 fF . The output pole is located at

$$f_{out} = \frac{1}{2\pi \cdot \frac{1}{g_{m2}} \cdot (C_{M2} + C_{sg2})} = \frac{1}{2\pi \cdot 6.5k \cdot (74 \text{ fF})} = 331 \text{ MHz} \quad (21.17)$$

The zero is located at

$$f_z = \frac{g_{m1}}{2\pi C_{gd1}} = \frac{150 \mu\text{A/V}}{2\pi \cdot 2 \text{ fF}} = 11.9 \text{ GHz}$$

The transfer function of the amplifier is then

$$A_v(f) = \frac{v_{out}(f)}{v_s(f)} = -\frac{\left(1 - j\frac{f}{11.9 \text{ GHz}}\right)}{\left(1 + j\frac{f}{58 \text{ MHz}}\right)\left(1 + j\frac{f}{331 \text{ MHz}}\right)}$$

Figure 21.9 shows the AC analysis simulation results for the amplifier of Fig. 21.8. What would happen if we used v_{in} instead of v_s ? We wouldn't see the input

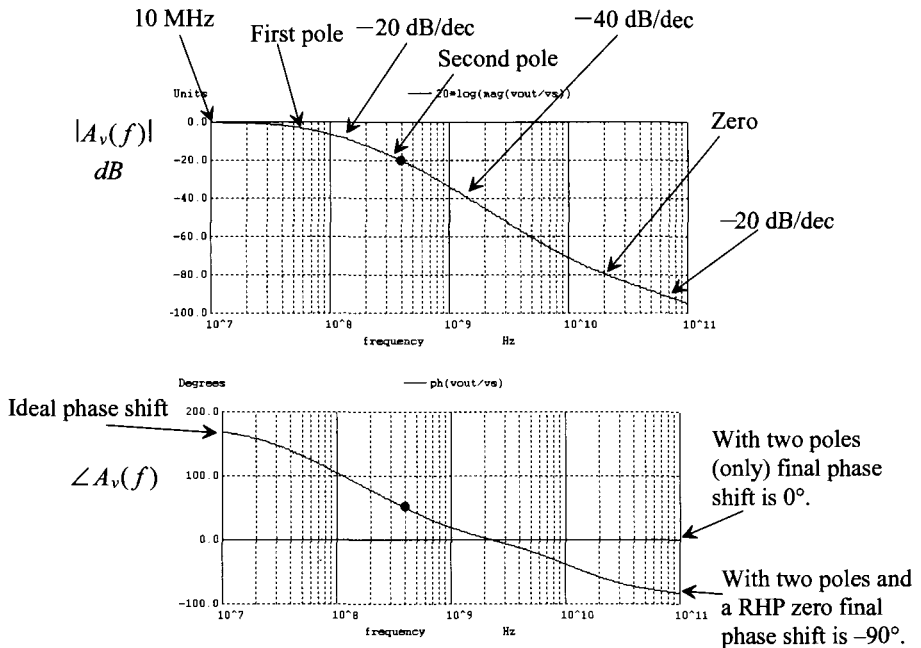


Figure 21.9 Frequency response of the amplifier in Fig. 21.8.

pole's effects on the frequency response. In the magnitude response we see that right after the first pole the gain decreases by a factor of 10 for every increase in frequency by 10 (equivalent to saying -20 dB/decade). We could also say that the gain decreases by 2 for every increase in frequency by 2 (equivalent to saying -6 dB/octave). After the second pole, the gain decreases by 100 for every increase in frequency by 10. The zero causes the frequency response to go back to -20 dB/decade. (The output and input of M1 are shorted together so that all we see are the effects of R_s charging the input capacitance of the amplifier.) It is useful to look at, and understand, the behavior of the amplifier from v_{in} to v_{out} in the simulation.

The effects of the RHP zero occur one decade before f_z . This *excess phase* shift is the problem we discussed earlier. It makes the amplifier appear not like an inverting amplifier (phase shift of -180 degrees) but rather like a noninverting amplifier sooner (at lower frequencies).

Note that magnitude of voltage gains in the neighborhood of 1 are common when using gate-drain-connected loads. At this point it may appear that this amplifier is almost worthless; however, we will see that when this amplifier is used with additional circuitry (e.g., a current source load for a differential pair), it can be very useful. ■

Example 21.3

Determine the magnitude and phase of the time-domain signal v_{out} , at 400 MHz, for the amplifier in Ex. 21.2 based on the frequency plots seen in Fig. 21.9. Verify the results using SPICE.

The magnitude response at 400 MHz is

$$\left| \frac{v_{out}}{v_s} \right| = -20 \text{ dB} = 0.1$$

and the phase response is

$$\angle \frac{v_{out}}{v_{in}} = 50^\circ = -310^\circ$$

The positive value means that the output is leading, or occurring earlier in time, than the input. Remembering that phase shift is simply an indication of a time delay at a particular frequency, we can write

$$50^\circ = 360 \cdot \frac{t_d}{T} = 360 \cdot t_d \cdot 400 \text{ MHz} \rightarrow t_d \approx 350 \text{ ps}$$

for a period of 2.5 ns. We'll use an input voltage of 1 mV to ensure small-signal operation. In the simulation that generated Fig. 21.9 our input AC signal was 1 V. Again, remember that SPICE replaces the active devices with linear models in an AC simulation so any value of amplitude can be used for the input signals. Using 1 V in an AC simulation is convenient because of the ratios often used when calculating gains. Figure 21.10 shows the SPICE simulation results. Notice how the peak amplitude of the input is 1 mV while the peak amplitude of the output (remembering the DC output voltage is $V_{DD} - V_{SG}$ or, from Table 9.1, approximately 3.85 V) is 0.1 mV (-20 db down from the low-frequency value). In

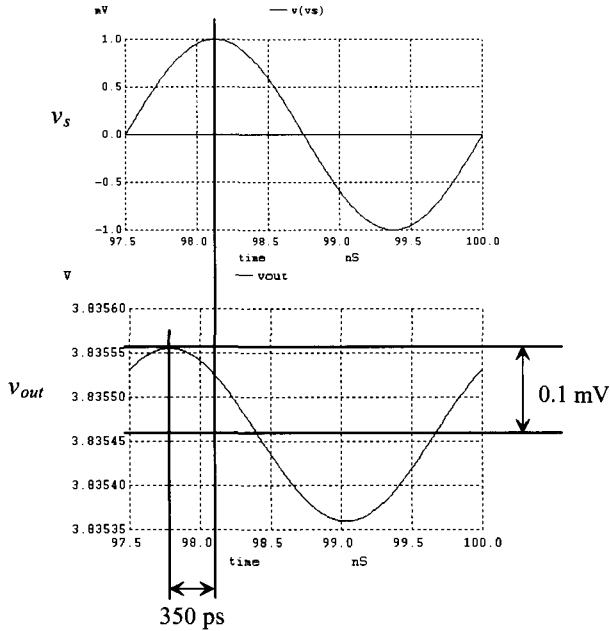


Figure 21.10 Time-domain behavior of the amplifier in Fig. 21.8 at 400 MHz.

a transient simulation we have to wait some time before the circuit settles to a stable operating point. In Fig. 21.10 we've waited 97.5 ns before looking at the output voltage. This gives the bias circuit, for example, time to turn on and generate the bias voltage needed in the amplifier (the value of the big capacitor was also reduced in this transient simulation so that its charging wouldn't increase the simulation time.) ■

A Common-Source Current Amplifier

The previous discussion was centered around CS voltage amplifiers. Next consider the circuit shown in Fig. 21.11. This amplifier can be thought of as a transimpedance amplifier, that is, voltage out and current in. The ideal input impedance of an amplifier with a current input is zero (all of the input current flows into the amplifier with this ideal input impedance). Note that the biasing of the amplifier is controlled by the current mirror formed between M3 and M1 (drawn in a different way to ensure that the reader recognizes the topology when it is seen).

The input impedance of the amplifier in Fig. 21.11 is given by

$$R_{in} = \frac{1}{g_{m3}} || r_{o3} || r_{o4} \approx \frac{1}{g_{m3}} \quad (21.18)$$

The input current, i_{in} , is determined by

$$i_{in} = g_{m3} v_{in} = g_{m3} v_{gs3} = g_{m3} v_{gs1} \quad (21.19)$$

The current through M1, and thus M2, is given by

$$i_d = g_{m1} v_{in} = \frac{g_{m1}}{g_{m3}} \cdot i_{in} \quad (21.20)$$

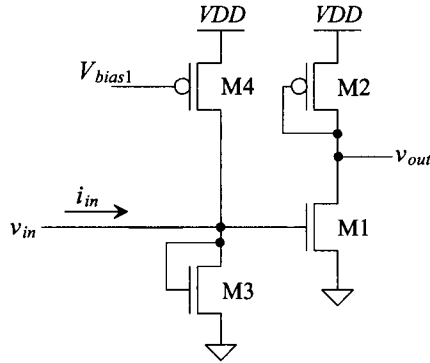


Figure 21.11 A transimpedance amplifier.

This equation can be rewritten, $g_{m3} = \beta_3(V_{GS3} - V_{THN})$ and $g_{m1} = \beta_1(V_{GS3} - V_{THN})$, as

$$\frac{i_d}{i_{in}} = \frac{\beta_1}{\beta_3} = \frac{W_1 L_3}{W_3 L_1} \quad (21.21)$$

That is, we can get a current gain simply by adjusting the size of M1 and M3. The current in M2 can then be mirrored out to a load. The transresistance gain of this configuration is given by

$$A_R = \frac{v_{out}}{i_{in}} = -\frac{i_d \frac{1}{g_{m2}}}{i_d \cdot \frac{g_{m3}}{g_{m1}}} = -\frac{g_{m1}}{g_{m2} g_{m3}} \quad (21.22)$$

Note that adding M3 and thus the resistor of value $\frac{1}{g_{m3}}$ to ground lowers the input time constant and increases the bandwidth of the amplifier.

Common-Source Amplifier with Source Degeneration

The amplifier in Fig. 21.3 is an example of a CS amplifier with source degeneration. The gain of the amplifier is reduced because the effective transconductance of the amplifier becomes

$$\frac{1}{g_{m,eff}} = \frac{1}{g_m} + R_s \quad (21.23)$$

as seen in Ex. 21.1. Adding the source resistance is useful, especially if $R_s \gg 1/g_m$, to precisely set the transconductance of an amplifier. Note that we didn't, in Ex. 21.1, include the resistance looking into the drain of M1 when we wrote the "resistance in the drain." *Let's review, or again verify, how to determine the small-signal, low-frequency resistances looking into the drain and source of a MOSFET.* Using this information will greatly aid in our intuitive analysis of CMOS circuits.

Consider the test circuit shown in Fig. 21.12. Here the DC voltages ensure that the MOSFET is operating in the saturation region. We are concerned with the AC current that flows through v_i . This test voltage divided by i_d gives us the small-signal AC resistance looking into the drain of the MOSFET. In other words, this is the resistance connected to ground at the drain node.

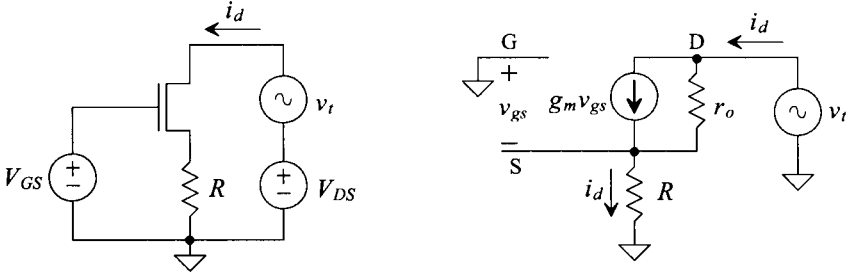


Figure 21.12 Circuit used to determine the resistance looking into the drain of a MOSFET.

Neglecting the body effect, we can write the test voltage as

$$v_t = (i_d - g_m v_{gs})r_o + i_d R \quad (21.24)$$

where (noting $v_g = 0$ and $v_s = i_d R$)

$$v_{gs} = -i_d R \quad (21.25)$$

The resistance looking into the drain of the MOSFET is then given by

$$R_o = r_d = \frac{v_t}{i_d} = (1 + g_m R)r_o + R \approx (1 + g_m R)r_o \quad (21.26)$$

keeping in mind that $r_o \approx \frac{1}{\lambda I_D}$. For a cascoded MOSFET topology ($R = r_o$), this is approximately, as seen in Eq. (20.53), $g_m r_o^2$. For a *triple cascode*, three MOSFETs in series, ($R = g_m r_o^2$), the output resistance is

$$R_o \approx g_m^2 r_o^3 \quad (21.27)$$

noting that the output resistance goes up with each added MOSFET by the open-circuit gain, $g_m r_o$.

Example 21.4

Repeat Ex. 21.1 without neglecting the loading effects of the output resistances of M1 and M2.

The resistance looking into the drain of M2 is $\frac{1}{g_{m2}} || r_{o2}$, while the resistance looking into the drain of M1 is $r_{o1}(1 + g_{m1}R_s)$. The exact gain of the circuit is

$$A_v = \frac{v_{out}}{v_{in}} = - \frac{\frac{1}{g_{m2}} || r_{o2} || [r_{o1}(1 + g_{m1}R_s)] || R_L}{R_s + \frac{1}{g_{m1}}}$$

or when $r_{o1} || r_{o2} \gg \frac{1}{g_{m2}}$, this reduces to

$$A_v = \frac{\frac{1}{g_{m2}} || R_L}{R_s + \frac{1}{g_{m1}}} \blacksquare$$

The resistance in the source of the MOSFET can be found using the circuit shown in Fig. 21.13. Neglecting the body effect and output resistance of the MOSFET gives

$$v_{in} = v_{gs} + i_d R \quad (21.28)$$

Since $g_m v_{gs} = i_d$, this equation can be written as

$$v_{in} = \frac{i_d}{g_m} + i_d R = i_d \left[\frac{1}{g_m} + R \right] \quad (21.29)$$

We can look at this in a simplified manner; that is, $\frac{1}{g_m}$ is the resistance looking into the source of a MOSFET, while R is the resistance connected from the source of the MOSFET to ground.

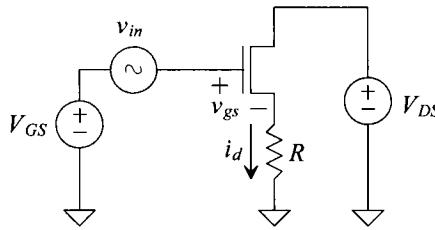


Figure 21.13 Circuit used to determine the resistance looking into the source of a MOSFET.

Noise Performance of the CS Amplifier with Gate-Drain Load

Figure 21.14 shows the CS amplifier of Fig. 21.1a with each MOSFET's model for noise power spectral density included (see Eq. [9.66]). Remembering that we only add the power from each noise source, we can write the output noise power spectral density as

$$V_{noise}^2(f) = \left(\frac{1}{g_{m2}} \right)^2 \cdot (I_{M1}^2 + I_{M2}^2) \quad (21.30)$$

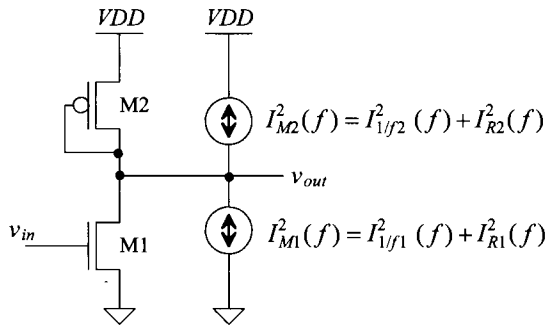


Figure 21.14 Noise modeling in a CS amplifier.

The input-referred noise power spectral density is then

$$V_{in}^2(f) = \frac{V_{noise}^2(f)}{A_v^2} = \left(\frac{g_{m2}}{g_{m1}}\right)^2 \cdot \left(\frac{1}{g_{m2}}\right)^2 \cdot (I_{M1}^2 + I_{M2}^2) \quad (21.31)$$

Minimizing the input-referred noise is accomplished by making g_{m1} large. Using the expressions in Eq. (9.66), we also see that minimizing g_{m2} reduces the input-referred noise. This is equivalent to saying large A_v results in minimum input-referred noise.

21.1.2 The Source Follower (Common-Drain Amplifier)

Source-follower configurations using a gate-drain-connected load are shown in Fig. 21.15. The drain of the amplifying device, that is, the MOSFET *not* gate-drain-connected (the load), is common to both the input and the output, so this topology is often called a common-drain amplifier. Setting the bias current in this topology can be challenging. In the following we don't concern ourselves with this issue.

A source follower implemented in CMOS has an asymmetric drive capability; that is, the ability of the follower to source current is not equal to the ability to sink current for a given bias condition and AC input signal. The small-signal gain of the NMOS source follower shown in Fig. 21.15 is simply determined by a voltage divider between the resistance looking into the source of M2 ($1/g_{m2}$) with the resistance of the gate-drain-connected load, M1 ($1/g_{m1}$). The gain of the **source follower amplifier** is

$$A_v = \frac{v_{out}}{v_{in}} = \frac{\text{resistance connected to the source}}{\text{resistance connected to the source} + \text{resistance looking into the source}} \quad (21.32)$$

or for the NMOS source follower in Fig. 21.15

$$v_{in} = \overbrace{v_{gs2}}^{i_d/g_{m2}} + \overbrace{v_{out}}^{i_d/g_{m1}} \quad \text{where } (v_{gs1} = v_{out}) \quad \text{and so } v_{out} = v_{in} \cdot \frac{\frac{1}{g_{m1}}}{\frac{1}{g_{m1}} + \frac{1}{g_{m2}}} \quad (21.33)$$

The output resistance of the source follower shown in this figure is

$$R_{out} = \frac{1}{g_{m1}} \parallel \frac{1}{g_{m2}} \quad (21.34)$$

Note that M2 can only source current, while M1 can only sink current, and the gain of the source follower is always less than one.

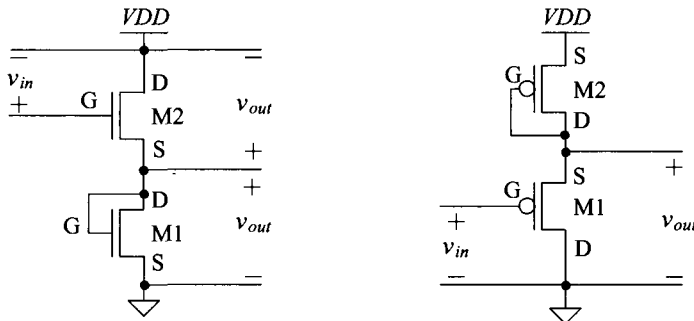


Figure 21.15 Source followers (common drain) using gate-drain loads.

21.1.3 Common Gate Amplifier

The common gate amplifier with gate-drain load is shown in Fig. 21.16. The input resistance of this amplifier is simply the resistance looking into the source of M1, or

$$R_{in} = \frac{1}{g_{m1}} \quad (21.35)$$

The gain of this **common-gate amplifier** is

$$A_v = \frac{v_{out}}{v_{in}} = \frac{-i_d \cdot \frac{1}{g_{m2}}}{-v_{gs1}} = \frac{-i_d \cdot \frac{1}{g_{m2}}}{-i_d \cdot \frac{1}{g_{m1}}} = \frac{\text{resistance in the drain}}{\text{resistance in the source}} = \frac{\frac{1}{g_{m2}}}{\frac{1}{g_{m1}}} \quad (21.36)$$

or noting the positive gain

$$A_v = \frac{g_{m1}}{g_{m2}} = \sqrt{\frac{KP_n}{KP_p} \cdot \frac{W_1 L_2}{W_2 L_1}} \quad (21.37)$$

which is the same form as the gain for the CS amplifier with gate-drain load.

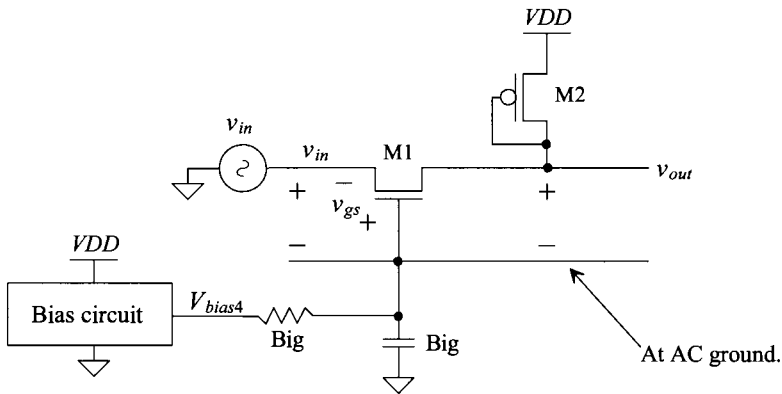


Figure 21.16 The common gate amplifier.

21.2 Current Source Loads

In the last section we talked about the gate-drain MOSFET part of a current mirror. In this section we talk about the current-source side of the mirror. The current source load provides an amplifier with the largest possible load resistance and thus the largest gain.

21.2.1 Common-Source Amplifier

Consider the CS amplifier with current source load shown in Fig. 21.17. The MOSFET M1 is the common-source component of the amplifier, while the MOSFET M2 is the current source load. The DC transfer characteristics are also seen in this figure. The amplifying portion of the curve occurs when the output is not close to V_{DD} or ground (both M1 and M2 are saturated). The slope of the line when both transistors are saturated corresponds to the gain of the amplifier.

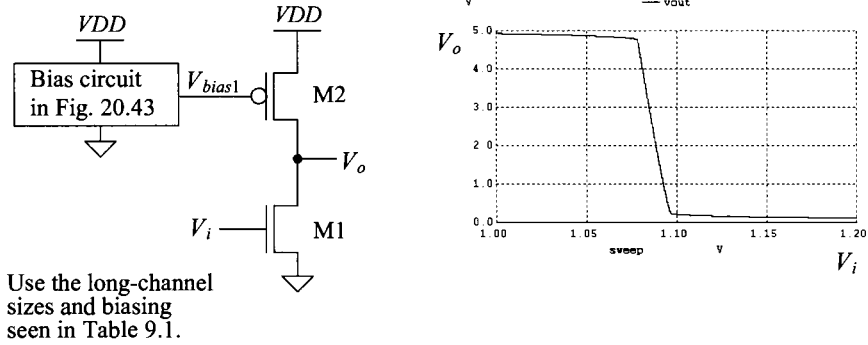


Figure 21.17 Common-source amplifier with current source load.

Class A Operation

We know from Table 9.1, and the bias circuit in Fig. 20.43, that M2 in Fig. 21.17 is biased to behave like a 20 μA current source. This means that if M1 shuts off, the maximum current we can supply to the output is 20 μA : this is important to understand. The CS amplifier in Fig. 21.17 is called a *class A* amplifier because, for proper operation, both MOSFETs are always conducting a current. In a *class B* amplifier, only one MOSFET is conducting a current at a given time. In *class AB* amplifiers (which we'll talk about later), both MOSFETs or a single MOSFET conduct a current at a given time.

Example 21.5

Estimate the maximum rate the load capacitor in Fig. 21.18 can be charged (estimate the slew rate across the load capacitor). Verify the estimate with SPICE.

The maximum current that the amplifier can source is 20 μA . The slew rate across the load is calculated using

$$I = C_L \cdot \frac{dV_{out}}{dt} \text{ or slew rate} = \frac{dV_{out}}{dt} = \frac{I}{C_L} \quad (21.38)$$

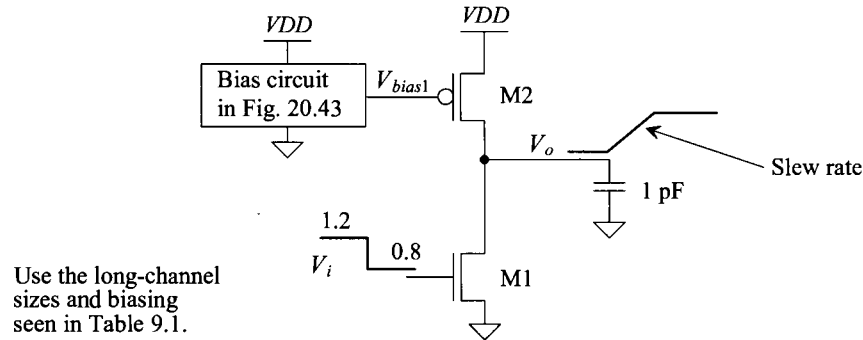


Figure 21.18 Slew rate limitations in a class A amplifier.

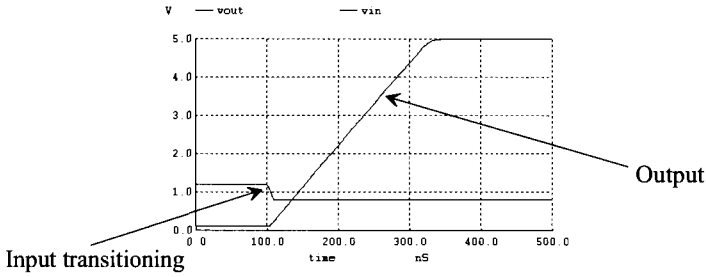


Figure 21.19 Verifying the results in Ex. 21.5

or

$$\frac{dV_{out}}{dt} = \frac{20 \mu A}{1 pF} = 20 V/\mu s$$

For the output to transition 5 V requires 250 ns. ■

Small-Signal Gain

If we follow the intuitive method discussed in the last section, Eq. (21.1), the voltage gain of a common-source amplifier is the total parallel resistance at the drain of M1 divided by the resistance in the source of M1, or

$$A_v = \frac{v_o}{v_i} = -\frac{r_{o1} || r_{o2}}{\frac{1}{g_{m1}}} = -g_{m1}(r_{o1} || r_{o2}) = -\frac{g_{m1}}{g_{o1} + g_{o2}} \quad (21.39)$$

where $r_{o1} = 1/g_{o1}$.

Open Circuit Gain

The gain of the amplifier can be increased by using a cascode current load in place of M2. The resistance looking into the drain of the cascode current source/load is much larger than the output resistance of M1. This situation is sometimes referred to as the *open circuit gain* of a common-source amplifier (see the specification in Tables 9.1 and 9.2). This specification indicates the maximum possible gain attainable with a single MOSFET (the MOSFET's load is its own output resistance r_o). The open circuit gain of M1 can be written as

$$\text{Open circuit gain} = -\frac{r_{o1}}{\frac{1}{g_{m1}}} = -g_{m1}r_{o1} = -\frac{\sqrt{2\beta_1 I_D}}{I_D \lambda_1} \quad (21.40)$$

High-Impedance and Low-Impedance Nodes

In Fig. 21.17, the output node at the drains of M1 and M2 is termed a *high-impedance node*, that is, a node with only drain connections. The effective resistance at this node to ground is $r_{o1} || r_{o2}$. A node connected to the source of a MOSFET or a MOSFET with its drain and gate connected is termed a *low-impedance node*. The small-signal resistance looking into the source of a MOSFET and the small-signal resistance of the gate-drain (diode)-connected MOSFET are both $1/g_m$. High-impedance nodes usually have a low-frequency pole associated with them that limits the speed of the amplifier.